



Security Assessment



Item 13 Internal Report

August 2026

Prepared for ether.fi



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Initial Commit Hash	Final Commit Hash
aa27177e	07342bdc

Protocol Scope

- src/top-up/TopUp.sol
- src/top-up/TopUpFactory.sol

Informational Issues

I-01. Permissionless `processTopUp` allows attackers to permanently grief the automated redirect flow for unsupported tokens

Description: The `TopUpFactory` exposes `processTopUp` and `processTopUpFromContracts` as permissionless external functions meant to sweep supported bridging tokens from users' `TopUp` contracts into the factory. However, these functions do not validate that the tokens passed in the `tokens` array are actually bridging-supported (i.e., they do not check `$.supportedTokens.contains(tokens[i])`).

This lack of validation allows anyone to call `processTopUp` with the address of an unsupported token that is intended for redirection (e.g., raw Backed xStocks). When called, the user's `TopUp` contract will blindly transfer these redirectable tokens into the `TopUpFactory`.

Once the tokens are maliciously swept into the `TopUpFactory`, the automated redirect flow is permanently broken for that user. When the backend attempts to call `batchRedirectToTradingSafe`, the `TopUp` contract will revert due to insufficient balance. Because the swept tokens are now commingled with other funds in the factory, the protocol administrators are forced to manually invoke `recoverFunds` to the `recoveryWallet` and perform off-chain forensics to trace the funds back to their rightful owners. This allows an attacker to cheaply and permanently DoS the system's automated recovery mechanism for all misrouted tokens.

Recommendation: Add validation in `TopUpFactory.processTopUp` and `processTopUpFromContracts` to ensure that only supported tokens can be swept. For example, add the following check inside the token processing loop (or before sweeping): `if (!$.supportedTokens.contains(tokens[i])) revert OnlySupportedTokens();`. Alternatively, perform this validation within `TopUp.processTopUp` by querying the factory's `isTokenSupported` mapping before executing the transfer.

Customer's response: Fixed in commit [07342bdc](#).

Fix review: Fix confirmed. The process top up functions now have a guard to ensure that only tokens supported for bridging are swept to the factory.

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